INCH-POUND

MIL-C-46935B (AR)
14 May 1990
SUPERSEDING
MIL-C-46935A (MU)
3 February 1966

# MILITARY SPECIFICATION

# CARTRIDGES, COMMERCIAL

This specification is approved for use by the U.S. Army, Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers commercial noncombat centerfire, rimfire, and shotgun cartridges of various calibers for use in rifle, pistol, revolver, and shotgun weapons.

1.2 Classification.

1.2.1 Type. This specification covers three types of commercial cartridges:

Type I - Centerfire

Type II - Rimfire

Type III - Shotgun

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1305

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-601 | Boxes, Wood, Cleated Plywood |
| PPP-B-621 | Boxes, Wood, Nailed and Lock-Cornered |
| PPP-B-636 | Box, Fiberboard |

MILITARY

| | |
|---|---|
| MIL-B-117 | Bag, Sleeve and Tubing - Interior Packaging |
| MIL-B-2427 | Box, Ammunition Packing: Wood, Nailed |
| MIL-B-46506 | Boxes, Ammunition Packing, Wood, Wirebound |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-109 | Quality Assurance Terms and Definitions |
| MIL-STD-129 | Marking for Shipment and Storage |

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.2 Non-Government publications. The follwing document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

SPORTING ARMS AND AMMUNITION MANUFACTURER'S INSTITUTE

Technical Committee Manual (S.A.A.M.I)

(Application for copies should be addressed to the Secretary, Sporting Arms and Ammunition Manufacturer's Institute, 250 E. 43rd Street, New York 17, NY).



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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The cartridge shall comply in order of precedence with the contract, applicable drawing, and the specification.

3.2 Performance. Requirements for the following characteristics shall be as specified in the S.A.A.M.I Manual.

3.2.1 Velocity. - Requirements for the velocity shall be as specified in the S.A.A.M.I. Manual.

3.2.2 Pressure. - Requirements for the pressure shall be as specified in the S.A.A.M.I. Manual.

3.2.3 Accuracy. - (for cartridge types I and II) Requirements for the accuracy shall be as specified in the S.A.A.M.I Manual.

3.2.4 Pattern. - (for cartridge type III) Requirements for the pattern shall be as specified in the S.A.A.M.I Manual.

3.2.5 Primer sensitivity. - Requirements for the primer sensitivity shall be as specified in the S.A.A.M.I Manual.

3.2.6 Function and casualty. - Requirements for the function and casualty shall be as specified in the S.A.A.M.I Manual.

3.2.7 Workmanship. Requirements shall be as specified in the S.A.A.M.I Manual, in referenced specifications, and in applicable drawings.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Quality Assurance terms and definitions. Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Inspection provisions.

4.2.1 Lot.

4.2.1.1 Size of Lot. The ammunition lot size shall be as agreed upon between the contractor and contracting officer.

4.2.1.2 Lot identification. Each lot of ammunmtion shall be identified as to type, caliber, and where applicable, model, as well as with an Army lot number and a Federal Stock Number when assigned by the procuring activity. No lot number shall be assigned to more than one lot of ammunition containing the same type and caliber cartridge or shot size.



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4.2.2 Examination. Cartridge examination shall be performed in accordance with the S.A.A.M.I. Manual. Visual and Dimensional quality shall conform to the levels specified therein.

4.2.3 Tests. The cartridge and primers shall be tested to determine compliance with requirements of 3.2 as applicable. Sample sizes and acceptance criteria shall be as specified in the S.A.A.M.I Manual.

4.2.4 Packaging,packing and marking inspection. Inspection for packaging, packing and marking shall be in accordance with established procedures for commercial ammunition to determine compliance with requirements of 5.

4.3 Inspection equipment. The contractor shall furnish and maintain all required measuring and testing equipment.

4.4 Methods of inspection. All tests shall be made in accordance with the methods and procedures and using the equipment specified in the S.A.A.M.I Manual.

4.5 Certifications of quality. For each lot presented for acceptance, the contractor shall furnish a certificate of compliance, certifying that the ammunition complies with all requirements of this specification and such other requirements as may be specified in the contract. Such certification shall include manufacturer's identification, results of examinations and tests made and shall be signed by a responsible agent of the contractor.

5. PACKAGING

5.1 Packaging. The cartridges shall be packaged in cartons in accordance with best commercial practice.

5.2.1 Level A. The quantity of cartridges specified shall be packed in fiberboard containers in accordance with Federal Specification PPP-B-636, class domestic. The fiberboard box shall be placed in a barrier bag made in accordance with MIL-B-117, Type I, Style 1, Class E, and heat sealed closed. The box shall then be overpacked in a treated wood box made in accordance with PPP-B-601, PPP-B-621, MIL-B-46506, or MIL-B-2427.

5.2.2 Level B. The quantity of cartridges specified shall be packed in a fiberboard container made in accordance with Federal Specification PPP-B-636, Class WWVR, Grade V3c or W5c, Style RSC, waterproofed and closed in accordance with PPP-B-636.



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5.2.3 Level C. The quantity of cartridges specified shall be packed in a fiberboard container made in accordance with Federal Specification PPP-B-636, Grade V3c or W5c, Style RSC and closed in accordance with Method V of PPP-B-636.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

5.4 Palletization. Palletization shall be as specified in the contract.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Intended use. The cartridges covered by this specification are intended for use in training, markmanship and service application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Nomenclature, drawing number, and Federal Stock Number of cartridge to be procured.

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. Provision for distribution of Certification of Quality.

e. Type and level of packing.

6.3 Technical Committee Manual (S.A.A.M.I). When requested by the prospective bidder,the contacting officer may obtain pertinent data from the Technical Committee Manual (S.A.A.M.I) upon application to the Secretary, Sporting Arms and Ammunition Manufacturers' Institute, 250 E. 43rd Street, New York, 17, NY.

6.4 Hazard notice. - The cartridges described herein and certain of their components are flammable and explosive or just explosive and consequently present hazards in manufacture, handling, storage, and shipment. The contractor should recognize these hazards and take appropriate measures to guard and protect against fire, explosion, adverse environment, corrosive atmosphere, rough handling and electrically induced incidents.



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6.5 Subject term (key word) listing.

Commercial
Centerfire
Rimfire
Shotgun

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - AR

Preparing activity
Army - AR

(Project 1305-AC58)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-C-46935B (AR) | 14 May 1990 |

**3. DOCUMENT TITLE**

CARTRIDGES, COMMERCIAL

**4. NATURE OF CHANGE** *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME | b. TELEPHONE (Include Area Code) | |
|---|---|---|
| US Army ARDEC<br>Standardization & Specification Office | (1) Commercial<br>(201) 724-6671 | (2) AUTOVON<br>880-6671 |
| c. ADDRESS (Include Zip Code)<br>ATTN: SMCAR-BAC-S/Bldg 6<br>Picatinny Arsenal, NJ 07806-5000 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |



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